Lattice Enthalpy and Born–Haber Cycles

Answer key with worked solutions

A. Definitions

1. Lattice enthalpy.

> (a) The enthalpy change when 1 mol of an ionic solid forms from its gaseous ions under standard conditions. (b) Lattice: $\mathrm{K^+(g) + Br^-(g) \longrightarrow KBr(s)}$. Formation: $\mathrm{K(s) + \tfrac{1}{2}Br_2(l) \longrightarrow KBr(s)}$ (bromine is a liquid in its standard state).

2. Endothermic or exothermic.

> (a) endothermic (b) **exothermic** (c) **endothermic** (d) endothermic (e) exothermic

B. Calculations

3. LiF.

> Steps to the gaseous ions:
>
> $$159.3 + 520.2 + 79.4 - 328.2\ (\mathrm{kJ/mol})$$
> $$= +430.7\ \mathrm{kJ/mol}$$
>
> $$\Delta H_{\mathrm{latt}}{}^{\circ}$$
> $$= -616.0\ \mathrm{kJ/mol} - 430.7\ \mathrm{kJ/mol}$$
> $$= \mathbf{-1046.7\ kJ/mol}$$

4. EA of bromine.

> $\Delta H_{\mathrm{f}}{}^{\circ} = \Delta H_{\mathrm{sub}}(\mathrm{K}) + \mathrm{IE}_1 + \Delta H_{\mathrm{atom}}(\mathrm{Br}) + \mathrm{EA} + \Delta H_{\mathrm{latt}}{}^{\circ}$, so:
>
> $$\mathrm{EA} = -393.8 - 89.0 - 418.8$$
> $$- 111.9 - (-689.0)\ (\mathrm{kJ/mol})$$
> $$= \mathbf{-324.5\ kJ/mol}$$

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C. Trends

5. Order.

> **KCl, NaCl, CaO, MgO** (least to most negative). KCl and NaCl have 1+ and 1− ions; K^+ is larger than Na^+, so KCl is less negative. CaO and MgO have 2+ and 2− ions, so their attractions are about four times stronger; Mg^{2+} is smaller than Ca^{2+}, so MgO is the most negative.

6. Second electron affinity of oxygen.

> (a) Steps other than EA2:
>
> $$\begin{aligned} &177.8 + 589.8 + 1145.4 \\ &\quad + 249.2 - 141.0 \ (\text{kJ/mol}) \\ &= +2021.2 \ \text{kJ/mol} \end{aligned}$$
>
> $$\begin{aligned} \text{EA}_2 &= -634.9 - 2021.2 \\ &\quad - (-3414) \ (\text{kJ/mol}) \\ &= \mathbf{+758 \ kJ/mol} \end{aligned}$$
>
> (b) It is **endothermic**: the electron is added to O^-, which is already negative, so it is repelled and energy must be put in.

D. Challenge

7. Hypothetical MgCl.

> (a)
>
> $$\begin{aligned} \Delta H_\text{f}^\circ &= 147.1 + 737.7 + 121.3 \\ &\quad - 348.6 - 753.0 \ (\text{kJ/mol}) \\ &= \mathbf{-95.5 \ kJ/mol} \end{aligned}$$
>
> (b) Products minus reactants:
>
> $$\begin{aligned} \Delta H &= (1 \ \text{mol})(-641.3 \ \text{kJ/mol}) \\ &\quad - (2 \ \text{mol})(-95.5 \ \text{kJ/mol}) \\ &= \mathbf{-450.3 \ kJ} \end{aligned}$$
>
> (c) MgCl would be stable relative to the elements, but it would release a lot of energy by turning into Mg and $MgCl_2$. Removing the second electron costs 1450.7 kJ/mol, but the much more negative lattice enthalpy of $MgCl_2$ (2+ ions) more than repays it, so $MgCl_2$ forms.

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8. Ionic model.

> NaCl is almost purely **ionic**: the model of separate spherical ions describes it well. In AgI, the large, easily polarized I^- ion is distorted by Ag^+, so some electron density is shared between them: the bonding has **covalent character**, which adds extra attraction not included in the ionic model.

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